

MATTHEW WATSON

University of Colorado Boulder
Department of Mathematics
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EDUCATION

University of Colorado Boulder

Ph.D., Mathematics (2026).

Advisor: Sebastian Casalaina-Martin.

Thesis: Singularities of Ball Quotients.

M.A., Mathematics, (2023).

University of Maryland

B.S., Mathematics, *magna cum laude* (2020).

TEACHING EXPERIENCE

University of Colorado Boulder

Course Assistant, MATH 1300 : Calculus I (Fall 2025).

Instructor of Record, MATH 2400 : Calculus III (Fall 2024, Fall 2023).

Instructor of Record, MATH 1300 : Calculus I (Spring 2024, Fall 2022, Summer 2022, Fall 2021).

Instructor of Record, MATH 2300 : Calculus II (Spring 2022).

Teaching Assistant, MATH 1300 : Calculus I (Spring 2025, Fall 2020).

Teaching Assistant, MATH 2300 : Calculus II (Spring 2021).

Teaching Assistant, MATH 2400 : Calculus III (Spring 2023).

Grader, MATH 2001 : Discrete Mathematics (Fall 2022).

Grader, MATH 3450 : Introduction to Complex Variables (Spring 2022).

Grader, MATH 4440 : Coding and Cryptography (Fall 2021).

University of Maryland

Strauss Teaching Assistant, MATH 141 : Calculus 2 (Spring 2020).

Strauss Teaching Assistant, MATH 140 : Calculus 1 (Fall 2019).

CREDENTIALS

Certificate in College Teaching, CU Boulder Center for Teaching and Learning, (expected Spring 2026).

MENTORING

Mentor, Math Department Peer Mentor Program, Fall 2024 – Spring 2025.

Algebra Mentor, Winter Prelim Program, Dec 2024.

SERVICE

Graduate Student Representative to the Graduate Committee, CU Boulder, Fall 2024 – present.

GRANTS, AWARDS

Burton W. Jones Teaching Excellence Award (Fall 2025)

CU Boulder Graduate School Summer Fellowship (Summer 2025).

Haller Fellowship (Summer 2024).

Natural Sciences Dean's Fellowship (Summer 2023).

Research Experience for Graduates Summer Fellowship (Summer 2021).

RESEARCH INTERESTS

My research interests are in algebraic geometry and complex geometry; more specifically, I am interested in the geometry of ball quotients, general type results, Hodge theory, and modular forms.

INVITED TALKS

2. *Singularities on Ball Quotients*. Colorado State University, FRAGMENT Seminar, December 11, 2025.
1. *Noncanonical Singularities on Arithmetic Ball Quotients*. CU Boulder, Front Range Algebraic Geometry Day, April 3, 2025.

SEMINAR TALKS

8. *On the Kodaira Dimension of Locally Symmetric Varieties II*. CU Boulder, Algebraic Geometry Seminar, October 16, 2025.
7. *On the Kodaira Dimension of Locally Symmetric Varieties I*. CU Boulder, Algebraic Geometry Seminar, October 9, 2025.
6. *Modular Forms and the Baily–Borel Theorem*. CU Boulder, Graduate Student Seminar, September 17, 2025.
5. *Introduction to D -modules*. CU Boulder, Algebraic Geometry Seminar, February 6, 2025.
4. *The Moduli Space of Complex Tori*. CU Boulder, Graduate Student Seminar, October 23, 2024.
3. *Du Val Surface Singularities*. CU Boulder, Algebraic Geometry Seminar, October 12, 2023.
2. *The Reid–Tai Criterion*. CU Boulder, Algebraic Geometry Seminar, November 3, 2022.
1. *Geometric Intuition Behind Algebraic Geometry*. CU Boulder, Graduate Student Seminar, November 2, 2022.

CONFERENCES, SCHOOLS, AND WORKSHOPS ATTENDED

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| Jul 2025 | Summer Research Institute in Algebraic Geometry, Colorado State University |
| Nov 2024 | Western Algebraic Geometry Symposium, University of Arizona |
| Nov 2024 | Moduli of Varieties, University of Utah |
| Jul 2024 | SL Math Summer School, Introduction to the Theory of Algebraic Curves, Berkeley, CA |
| Nov 2023 | Western Algebraic Geometry Symposium, Washington University in St. Louis |
| Apr 2023 | Western Algebraic Geometry Symposium, University of Washington |
| Nov 2022 | Western Algebraic Geometry Symposium, UC Riverside |
| Apr 2022 | Western Algebraic Geometry Symposium, Colorado State University |